

Eshaan Saini

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EDUCATION

Texas A&M University, College Station, TX

Aug 2023 – Dec 2026

B.S. Computer Science, Minor in Cyber Security

GPA: 4.0 and **Dean's Honor Roll** (4 semesters)

Relevant Coursework: Data Structures, Operating Systems, Analysis of Algorithms, Cloud Computing, Databases, Software Engineering, Computer and Network Security

TECHNICAL SKILLS AND CERTIFICATIONS

- **Python, Java, C++, TypeScript, Go**, JavaScript, SQL, R, Haskell, React, Next.js, Angular, Node.js, Express, FastAPI, Flask, PyTorch, scikit-learn, Tailwind CSS, Git, Docker, AWS (RDS, S3, Cognito, EC2), PostgreSQL, MongoDB, Cypress, Linux, Nginx
- Building AI Agents with RAG and LangChain by IBM; Model Context Protocol by Anthropic; Machine Learning by Andrew Ng Stanford, Python for Data Science, AI and Development by IBM

EXPERIENCE

IBM – *Backend Development Intern*, Austin, TX

May 2026 – Aug 2026

- Building an **MCP server** in **Go** over the **IBM Cloud Logs API** spanning all **13 geo-distributed PowerVS** log environments (12 production regions plus staging); collapses a cross-region investigation that previously required 13 separately authenticated API calls into a single tool call, retrieving logs roughly **3x faster**
- Developing an **AI-native diagnostic agent** that runs log queries through the MCP server and generates structured **root-cause analysis** reports, cutting Level 3 incident triage time **50%** versus manual evidence gathering
- Designing a **RAG** knowledge base over anonymized historical support cases so the agent surfaces similar past incidents during live diagnosis

H-E-B – *Software Development Intern*, San Antonio, TX

June 2025 – Aug 2025

- Created **8 Cypress** automated end-to-end tests covering counterfeit coupon workflows (create, update, delete) in the H-E-B Coupon Creation modal, reducing manual testing time before deployment
- Modified **Angular** frontend components to fix bugs in the Counterfeit Coupons tab and built **REST API** endpoints for secure, efficient data access during testing workflows
- Spearheaded an overhaul of the engineering onboarding process by auditing docs, removing redundant steps, and producing updated documentation, leading to a **50% increase** in onboarding efficiency

Vector Edge – *Software Development Intern*, Austin, TX

June 2024 – July 2024

- Implemented **Google RE2** regex and the **Aho-Corasick Algorithm** within a classification SDK to identify **PII, PFI, and PHI** across large text datasets, matching every sensitive-data pattern in a single linear pass
- Built a context-based evaluation layer that assigns confidence scores to sensitive-keyword matches, improving detection accuracy by over **40%**
- Contributed to porting the SDK from **Windows** to **macOS/Linux**, expanding the addressable client base by **10%**

Texas A&M University – *Peer Teacher*, College Station, TX

Aug 2025 – Present

- Lead weekly lab sessions for **30+ students** in **Honors Computer Architecture** and Computer Systems, holding office hours for individualized support on low-level concepts
- Tutor students 1-on-1 in **Data Structures**, covering trees, graphs, hash maps, and algorithm complexity; collaborate with faculty to align lab content with lecture material

PROJECTS — (github.com/eshaansaini2004)

- **Stackbox:** Built a commercial real estate intelligence platform (**Next.js, FastAPI, PostgreSQL, AWS**) with **25+** REST endpoints for full CRUD on buildings, tenants, and occupancies; implemented **AWS Cognito** JWT auth middleware and **Docker Compose** deployment for production-grade reliability
- **Prompt Injection Firewall:** Built a drop-in **OpenAI-compatible** reverse proxy that detects and blocks prompt injection attacks via a two-layer AI-native detection engine (heuristic pass under **0.5ms**, semantic embedding fallback); covers **14 attack categories** from the OWASP LLM Top 10, validated by **41 automated tests**, with a live **WebSocket** monitoring dashboard
- **ShareTea POS:** Led a **5-member** team to build and deploy a full-stack point-of-sale system in **under 4 weeks**; **AWS**-hosted **PostgreSQL** database with **90,000+** entries, role-based access, and a self-hosted production deployment pipeline via **Nginx** reverse proxy and **pm2** process monitoring
- **Code Cartographer:** Built an AI-powered codebase analysis tool (**Python, Flask, Google Gemini API**) at HackTX 2025 with a real-time **WebSocket** dashboard processing **50+ files**, interactive dependency graphs, and a **VS Code extension** for in-editor AI-assisted analysis
- **LeopardWorks:** Built a **Discord** bot adopted by **30–35 TAMU students** for course planning; `/lookup` aggregates professor GPA distributions and live section availability from multiple scraped sources, while `/select` automates **Howdy** registration via headless **Playwright** with **Duo MFA** support